

Advanced Quantum Optics:

Quantum Emitter simulation with FDTD

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Abstract

We will conduct in this paper a brief report of a practical work about the simulation of a Quantum Emitter (QE) in various environments using FDTD software named RSoft CAD EnvironmentTM. This work is firstly made as a first touch about FDTD simulation and QE study for M2 Physics students.

The work consists in three different parts with different environment, firstly in Vacuum, then in air with a rectangular cuboid (PEC or gold material) interaction and lastly with (golden) plasmonic nano-antenna interaction.

This work involves multiple Python codes in order to treat the different data from simulations.

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1 Introduction

This report presents a brief study of a Quantum Emitter (QE) simulated in different electromagnetic environments using FDTD method implemented in *RSoft CAD Environment*TM. The objective is to observe how the local environment affects the radiated power, decay rate and quantum yield of the emitter.

Starting from a reference case in vacuum, we progressively introduce interactions with a Perfect Electric Conductor (PEC), a gold surface, and plasmonic nanoantennas in order to visualize how material properties and geometry modify the emission characteristics.

This work aims to provide a first contact with quantum optics simulation tools and to connect it with numerical observations, using advanced quantum physical concepts: the Local Density of States (LDOS).

Please, feel free to look at the Python codes provided in order to link it with the following analyses. The respective codes for each part is specified in the following section.

2 Material & Methods

2.1 Software and units system

Using the software RSoft CAD Environment™ we need to define the Quantum Emitter (QE) and its physical properties/environment before launching any simulation.

The different parameters are:

- Emission wavelength: 800 nm;
- Computation window: $2 \times 2 \times 2 \mu m^3$;
- Dipole at (0, 0, 0) the center of the window that is oriented differently through parts;
- Meshgrid of 10 nm cell along each direction;
- Time limit for the simulation: 6 μs .

The software uses a non-SI units set and we need to transform data in order to process them in calculations. To do so, we know that $\varepsilon_0 \approx 8.854 \times 10^{-12}$ SI and $\mu_0 \approx 4 \pi \times 10^{-7}$ SI and we can thus build the impedance $Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}$ and rebuild correct data with it.

For example, the values of magnetic field H_{comp} need to be divided by Z_0 in order to be in $A.m^{-1}$ (and multiplied by 10^6).

2.2 Desired values

For the computations, we need some theoretical formula:

$$\mathcal{P} = \oint_S \langle \vec{\Pi} \rangle \cdot d\vec{S} \quad (1)$$

$$= \oint_S \frac{\Re\{\vec{E}_0 \wedge \vec{H}_0\}}{2} \cdot d\vec{S} \quad (2)$$

$$= \frac{\omega}{2} \Im\{\vec{p}^* \cdot \vec{E}\} \quad (3)$$

$$\vec{p} = p e^{-i(\omega t + \varphi_p)} \vec{u}_p \quad (4)$$

$$\mathcal{P}_0(\text{vacuum}) = \frac{p^2}{12\pi} \times \frac{\omega}{\varepsilon \varepsilon_0} k^3; \quad k = \frac{2\pi}{\lambda} = \frac{\omega}{c} \quad (5)$$

$$\Leftrightarrow p = \sqrt{\frac{12\pi\varepsilon\varepsilon_0}{\omega k^3}} \mathcal{P}_0 = \sqrt{\frac{12\pi\varepsilon\varepsilon_0}{c k^4}} \mathcal{P}_0 \quad (6)$$

$$\varphi_p = \varphi_{E_0} + \arcsin\left(\frac{2\mathcal{P}}{\omega E_0 p}\right) \quad (7)$$

We can briefly prove (7):

$$\mathcal{P} = \frac{\omega}{2} \Im\{\vec{p}^* \cdot \vec{E}\} \quad (8)$$

$$= \frac{\omega}{2} \Im\{p E_0 e^{i(\omega t + \varphi_p)} e^{-i(\omega t + \varphi_{E_0})}\}$$

$$= \frac{\omega p E_0}{2} \Im\{e^{i(\varphi_p - \varphi_{E_0})}\}$$

$$\mathcal{P} = \frac{\omega p E_0}{2} \times \sin(\varphi_p - \varphi_{E_0}) \quad (9)$$

$$\Leftrightarrow \sin(\varphi_p - \varphi_{E_0}) = \frac{2\mathcal{P}}{\omega p E_0}$$

$$\varphi_p = \varphi_{E_0} + \arcsin\left(\frac{2\mathcal{P}}{\omega E_0 p}\right)$$

□

By the way, an important value is the phase shift $\Delta\varphi = \varphi_p - \varphi_{E_0}$ which is the only physically relevant data (because it is not simulation-dependant).

Note that the goal will be first to calculate the power from (2) on each face of the QE and then to build the dipole moment $\vec{p}$ (norme first then phase) all in the vacuum.

We can then work with surrounding matter to observe different properties and use the quantum relation:

$$\frac{P}{P_0} = \frac{\gamma}{\gamma_0} \quad (10)$$

$$\left\{ \frac{P_{\text{tot}}}{P_0} = \frac{P_{\text{rad}}}{P_0} + \frac{P_{\text{abs}}}{P_0} \right\} \equiv \left\{ \frac{\gamma}{\gamma_0} = \frac{\gamma_{\text{rad}}}{\gamma_0} + \frac{\gamma_{\text{loss}}}{\gamma_0} \right\} \quad (11)$$

2.3 Programs details and repository structure

We want to end this section by explaining the structure of files and programs used throughout this study.

The simulation can be found in folders named “VacuumDATA/“, “PECDATA”, “AuDATA”, ... and ...

They contain raw datafiles from the software and are treated in python files (names are related to the folders).

The core algorithm was made with the Vacuum case (simplest) and is slightly modified for the other cases.

Please for the next results, follow the Python codes provided for details, this study will not go deep into the code in order to, willingly, shorten this paper.

3 Results

This part will present results and answer some guiding questions to understand better QE-related phenomena and FDTD software manipulation.

3.1 Vacuum - no material

The FDTD measure is made on a simulation by using 6 monitors all around the QE in order to get the following values with the program `Vacuum.py`:

```
Xbot: P = 0.33 mW
Xtop: P = 0.33 mW
Ybot: P = 0.93 mW
Ytop: P = 0.93 mW
Zbot: P = 0.93 mW
Ztop: P = 0.93 mW

Total Power: Ptot = 4.398 mW
Norme of dipole moment: 1.13e-24 C m

Angle of dipole moment: 9.02°
The other theoretical value is 188.97°

Reconstruction with first angle 9.02°: 4.398 mW == Simulation 4.398 mW
Reconstruction with second angle 188.97°: 4.398 mW == Simulation 4.398 mW
Dephasage: 2.30e-02° or 1.80e+02°
```

We build the power on each surface by using eq. (2) paying attention to $\vec{d^2S}$ orientation depending on the surface (should point outside of the structure) and recalling that the meshgrid unit is a cell of $10 \times 10 \text{ nm}^2$.

Summing on each surface gives the Total Power Emitted by the QE. In the vacuum this will be called $\mathcal{P}_0$ because there are no subsidiary effects.

Then using eq. (6) we can deduce the norme of the dipole moment: p .

Nota bene: here, in the formula, $\varepsilon = 1$ due to vacuum.

Finally, we can obtain the absolute phase of $\vec{p}$ or the phase shift $\Delta\varphi$ from eq. (7) but another monitor needs to be used in order to obtain the electric field value AT QE coordinates: E_0 .

This E_0 -field is vectorially oriented in the same way as the QE (which oscillating is linearly), so we need to pay attention to this in the simulation.

Two possible values for φ_p :

we know from eq. (7) that the phase is obtained from an `arcsinus`, but mathematically, one `arcsin` value is the image of 2 angles in the unit circle! The `arcsin` function is obviouslyly bijective but only in the segment $]-\pi/2; \pi/2[$, so other solutions can be obtained by modulus and the unit circle is defined on $[0; 2\pi[$ so two values exist {at least} between 0-360° linked by a π rad translation.

Simplest example could be $\sin(\pi) = \sin(0) = 0$.

Finally, we can test to rebuild the power by using eq. (9) and hopefully recover the same value.

We physically expect a phase shift very low between the electric field and the dipole moment, which is verified here for the first phase value. This way, we can chose which phase value is the most relevant with respect to phase shift!

We can conclude by noting, for the next cases that in the vacuum we found a power of ≈ 4.398 mW.

3.2 Perfect Electrical Conductor (PEC) cases

A PEC is a material that acts like a mirror for waves and we expect to see this kind of phenomenon throughout two examples.

3.2.1 QE perpendicular to the PEC surface

We set the simulation such that the QE dipole moment is oriented along $\vec{e}_z$ while the PEC is a surface covering the xOy plan. Actually we set a distance of 20 nm between QE and PEC surface.

The program `PEC.py` gives:

Radiated Power: Prad = 8.676 mW

Angle of dipole moment: 90.00°

The other theoretical value is 269.95°

Reconstruction with first angle 90.00°: 8.689 mW == Simulation 8.676 mW

Reconstruction with second angle 269.95°: -0.384 mW != Simulation 8.676 mW

Dephasage: 4.81e-02° or 1.80e+02°

Total power: Ptot = 8.689 mW

$\gamma_r/\gamma_0 = 1.97$

$\gamma/\gamma_0 = 1.98$

Quantum yield is ~ 1.00

In this simulation, we know that theoretically two angles are possible mathematically to build the Total Power P_{tot} but only one coincides with the value we had in Vacuum. We thus will select this angle as the right one throughout the next programs.

N.B. we can also take notice that this angle is also the one for which the phase shift (between dipole moment and electric field phases) is the lowest. This result reinforces the choice of angle.

Also, the Power Radiated $P_{\text{rad}} = P_{\text{tot}}$ in this case which is completely relevant and expected because the PEC should not absorb the radiation from QE! We can also notice that the power is twice larger than in vacuum. We will discuss this further in Discussion 4.

We use relation (10) as such:

$$\begin{cases} \frac{P_{\text{rad}}}{P_0} = \frac{\gamma_r}{\gamma_0} \\ \frac{P_{\text{tot}}}{P_0} = \frac{\gamma}{\gamma_0} \end{cases}$$

And we find almost the same results. We can deduce from (11) that $\frac{P_{\text{loss}}}{P_0} = \frac{P_{\text{tot}} - P_{\text{rad}}}{P_0} = 0$ in this case. As said above, this is expected because PEC is a perfectly non-absorbing material.

Finally, the quantum yield can be defined as:

$$q = \frac{\gamma_r}{\gamma} = \frac{\gamma_r}{\gamma_0} / \frac{\gamma}{\gamma_0}$$

and in this case is obviously equal to one, $q = 1$.

3.2.2 QE parallel to the PEC surface

We use the same settings as before but we only modify the orientation of the dipole moment (of the QE) to be parallel to the PEC surface.

The program `PEC_ex.py` gives:

Radiated Power: $P_{\text{rad}} = 86.232 \mu\text{W}$

Angle of dipole moment: 90.00°

Phase shift: $6.77\text{e-}04^\circ$

Total power: $P_{\text{tot}} = 126.791 \mu\text{W}$

$\gamma_{\text{r}}/\gamma_{\text{0}} = 0.02$

$\gamma/\gamma_{\text{0}} = 0.03$

Quantum yield is ~ 0.68

So, in this case, the most important information is the fact the total and radiated powers are extremely low (1,000 times) compared to previous case. This means the PEC destroys some electromagnetic waves? Does it lock the quantum field in a more restricted way? We will also answer that in Discussion 4.

Also, we find in this case very low values of relative decay of radiated/total power ($\gamma_{\text{r}}/\gamma_{\text{0}}$ and γ/γ_{0}) but we can still try to compute the quantum yield and as expected, it is not one anymore. The yield is of the order of 68% which means that the system loses some of its energy in a way or another that cannot be retrieved by the user. The computed loss is $40.559 \mu\text{W}$ that vanishes.

The conclusion we can deduce from these two last experiments is that the orientation of the QE greatly impacts the way we can interact with the system and should be considered experimentally with care. Moreover, if we expect to use a non perfect mirror (which is the only thing possible in experiments) we need to take care of the absorption of the material AND this possible destruction of energy due to orientation. Finally, for obvious reasons, the best orientation is to set the dipole moment perpendicular to the surface of surrounding material in order to have maximum power.

3.3 Golden surface

We take exactly the same settings of simulation as the first PEC experiment, see 3.2.1, but we just change the kind of material used, here Au gold, which changes the simulation due to different properties of this metal. We know that gold is an excellent electrical conductor but it still absorbs some EM radiation as any material.

The program Au.py gives:

Radiated Power: Prad = 13.44 mW

Angle of dipole moment: 90.00°

Phase shift: 1.15e-01°

Total power: Ptot = 20.69 mW

$\gamma_r/\gamma_0 = 3.06$

$\gamma/\gamma_0 = 4.70$

Quantum yield is ~ 0.65

We see that the right orientation of QE allows us to recover a decent power radiated of the order of 10-20 mW. The relative decays values are 3 to 4 times greater than 1 which means the decays are far more high than the vacuum reference. Finally, the quantum yield is approximately of the order of the parallel PEC with 65%. The loss of ≈ 7 mW is purely absorbed by the gold surface in this case, knowing it is a pretty good conductor, we can't expect far better for a metallic surface experimentally and thus need to consider around 35% loss.

3.4 QE with golden nanoantenna

We set at 15nm from the dipole one or two golden antenna which are simulated as spheres of 170nm diameter.

3.4.1 One antenna

Remark: in this case we set the dipole moment along the diameter of the sphere (which is an analogy with the perpendicular PEC case).

The program named `Antenum.py` outputs:

Radiated Power: $P_{rad} = 35.05 \text{ mW}$

Angle of dipole moment: 90.00°

Phase shift: $2.29e-01^\circ$

Total power: $P_{tot} = 41.19 \text{ mW}$

$\gamma_r/\gamma_0 = 7.97$

$\gamma/\gamma_0 = 9.37$

Quantum yield is ~ 0.85

These results are quite suprising at first because we see that the sphere of gold increases the total and radiated power of the QE around the order of 35-40 mW; thus, the quantum relative decays are far more important and we have a quantum yield of 85% which is more than with a gold surface.

This means the user, “connecting” the golden sphere and using it as a receptor, can access to more power and with less losses. Here the loss is $\approx 6\text{mW}$ but it gives a ratio of 15% only!

We indeed understand that experimentally, this kind of setup will be the most interesting one because we need to use less (expensive) metal for better results.

3.4.2 Two antennas

In this simulation we will dispose two spheres around the quantum emitter, which dipole moment is oriented along the line passing through both spheres' centres.

The program `Antenae.py` provides:

Radiated Power: $P_{rad} = 235.78 \text{ mW}$

Angle of dipole moment: 90.00°

Phase shift: $1.55e+00^\circ$

Total power: $P_{tot} = 257.66 \text{ mW}$

$\gamma_r/\gamma_0 = 53.61$

$\gamma/\gamma_0 = 58.59$

Quantum yield is ~ 0.92

Suprisingly, in this case the power is demultiplicated and we find high relative decay rates/ power either total or radiated for the lowest losses (8%). We can think about it and see it as a resonance between both antennas, which is called Purcell Effect. This is finally the most interesting case that should be used during experiments in order to recover the most results with amplified powers, which is important for low values of power in order to be detected.

4 Discussion

The variations observed in radiated and total powers can be interpreted through the concept of the **Local Density of States (LDOS)**. The LDOS, noted $\rho(\vec{r}, \omega)$, represents the number of available electromagnetic modes per unit volume and frequency at the emitter position $\vec{r}$. It controls the spontaneous emission rate Γ :

$$\Gamma(\vec{r}, \omega) = \frac{\pi \omega}{3 \hbar \varepsilon_0} |\vec{p}|^2 \rho(\vec{r}, \omega) \quad (12)$$

Thus, any modification of the environment that alters $\rho(\vec{r}, \omega)$ affects directly the emission rate and thus the power and quantum yield of the QE!

- Vacuum case:

In the free space, the LDOS is uniform and isotropic, leading to a reference emission rate Γ_0 and power P_0 .

- PEC surface:

For a dipole perpendicular to a “perfect mirror”, the field reflected by the PEC interferes (constructively) with the QE dipole field. This doubles the LDOS near the emitter and we find something like $\rho/\rho_0 \approx 2$. This explains the observed factor 2 in P_r and the unity quantum yield ($q \approx 1$) expliciting the non-absorption phenomenon using PEC.

Conversely, when the dipole is parallel to the PEC, there is destructive interferences that cancel local modes leading to extremely weak emission of the QE.

- Gold case:

Here the material is absorbant which is seen with a quantum yield around 0.65. The LDOS description here can be seen as the Joule effect heating (from absorption) that corresponds to excitation of non-radiative LDOS modes.

- Nanoantennas:

These kind of contraption creates a strong local field confinement and resonance phenomenon of the LDOS. With one antenna, the LDOS and emission rate increase by one order of magnitude; with two antennas, the coupling between LDOS modes produces a resonant cavity boosting effectively the emission.

Overall, the results confirm that the spontaneous emission rate of a QSE is not an intrinsic property of the emitter but depends highly on the configuration of its surrounding. The goal can be seen as the optimization of the environment in order to boost the LDOS (mirros structures, resonant cavities, etc.) and try to reduce absorption (depends on the material used) and reduce destructive interference. wri

5 Conclusion

All the results we described in 3. Results are summed up in the following table:

cases \ values of interest	P_{rad}	P_{loss}	P_{tot}	γ_r/γ_0	γ/γ_0	q
Vacuum	4.40 mW	0	4.40 mW	N/A	N/A	N/A
PEC $\perp$ surface	8.68 mW	13.0 μ W	8.69 mW	1.97	1.98	100%
PEC $//$ surface	86.23 μ W	40.56 μ W	126.79 μ W	0.02	0.03	68%
Au surface	13.44 mW	7.25 mW	20.69 mW	3.06	4.70	65%
Au antenna	35.05 mW	6.14 mW	41.19 mW	7.97	9.37	85%
2 Au antennas	235.78 mW	21.88 mW	257.66 mW	53.61	58.59	92%

Table. Summing up all results of simulations

Through different simulated environments used in this paper, we have shown that the emission of a QE depends strongly on its surroundings. The PEC case highlighted interference effects and orientation dependence between QE dipole and the environment. The gold surface however introduced absorption losses and the nanoantennas, by contrast, produced a strong enhancement of the radiated power (which can be explained with LDOS amplification/coupling).

These results confirm that the spontaneous emission rate is not dependant solely on the QE but is dependant also on the electromagnetic environment. Optimizing the environment – by controlling materials, distances, orientations, etc. – is the key to improve light-matter interaction in practical quantum / nanophotonic systems.

Bibliography

The following references have been used to apprehend the subject and where skimmed through and not very well developed, to barely have the keys to understand how to treat this subject.

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